DecoR: deconfounding time series with robust regression [2]

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Keywords: Causal Inference · Robust Regression · Time Series Analysis

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion

Importance of Deconfounding - Earth Science

some time series figure of air pressure, precipitation, greenhouse gas?

Importance of Decounfounding - Earth Science

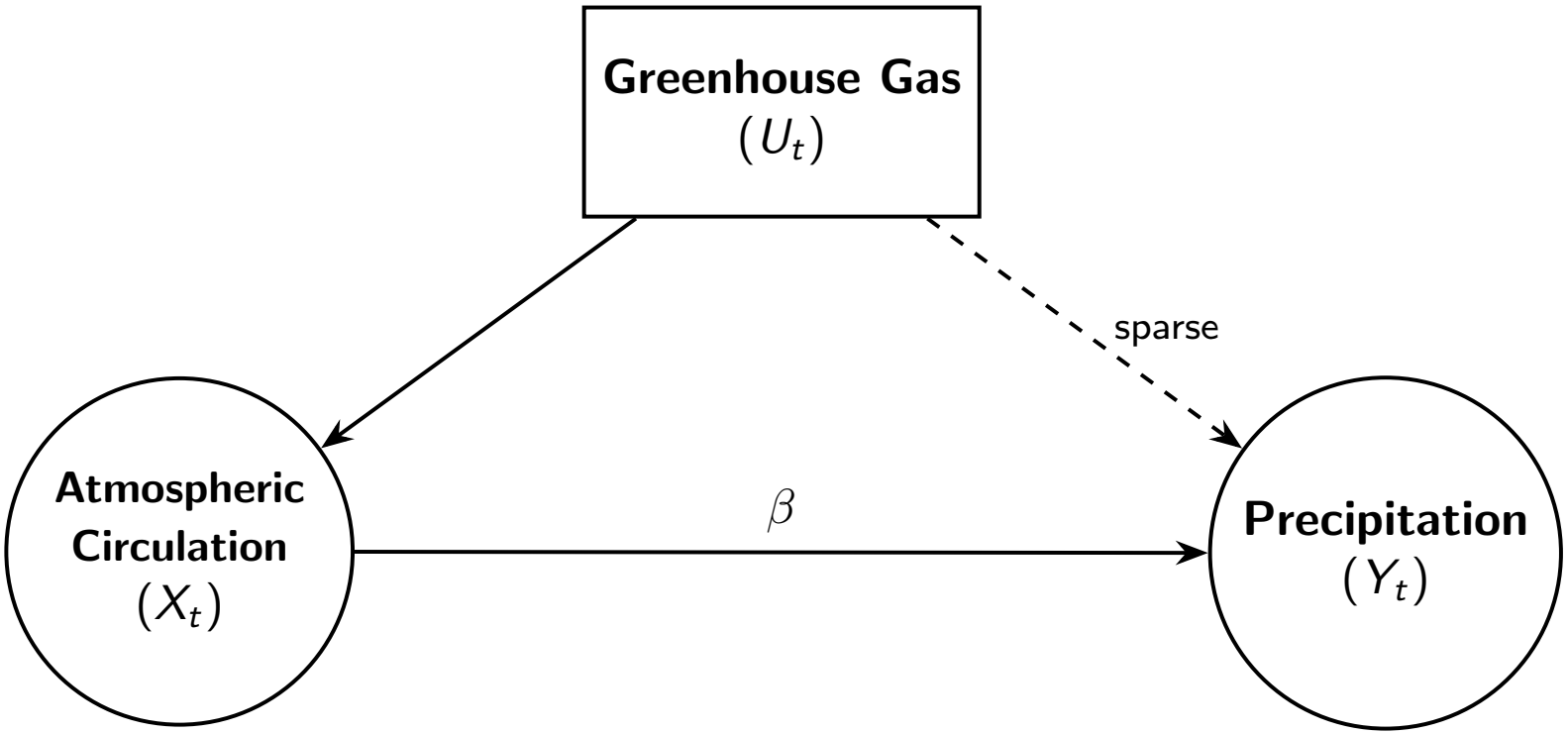

Figure: Causal Graph of Greenhouse Gas Confounding [3]

- **Goal:** Estimating β (effect of X to Y)
- **Problem:** Biased estimation of β (via standard OLS) without considering hidden confounder U
- **Idea:** Removal of confounder U seems important for accurate estimation of β

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The Underlying Model

Setting 1: Data Generating Process

Let $T \in \mathbb{R}_{>0}$ be a fixed time horizon and $d \in \mathbb{N}$. We consider the relationship:

$$Y_t = X_t^\top \beta + U_t + \eta_t, \quad t \in [0, T] \tag{1}$$

Components:

- $X_t \in \mathbb{R}^d$: Observed multivariate predictors.
- $U_t \in \mathbb{R}$: Unobserved hidden confounder.
- η_t: i.i.d. centered Gaussian noise.
- $\beta \in \mathbb{R}^d$: True causal effect (estimand).

Assumptions and Sparsity

Definition 1: (ϕ, G)-sparse process [cite: 164]

A process U is (ϕ, G)-sparse with respect to basis ϕ if for a set $G \subseteq \mathbb{N}$:

$$\langle U, \phi_k \rangle_{L^2} = \int_0^T U_t \phi_k(t) dt = 0 \quad \text{for all } k \notin G \text{ almost surely}$$

Assumption 1: Sparse Confounding

The set G is "small," meaning the hidden confounder only affects a few basis components (frequencies).

Implication: By transforming the data, we can isolate the "clean" frequencies G^c where the confounder disappears.

Example: Sparse U

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import dct

n = 500
T = 1.0  # Time horizon
t = np.linspace(0, T, n, endpoint=False)

# some small frequency set
k1, k2 = 2, 5

U_time = 2.0 * np.cos(2 * np.pi * k1 * t / T) + 1.0 * np.cos(2 * np
    .pi * k2 * t / T)

U_freq = dct(U_time, type=2, norm='ortho')
k_indices = np.arange(len(U_freq))
magnitude = np.abs(U_freq) / (n / 2) # Normalize

# 'magnitude' is zero everywhere except
# exactly at the integer indices k=2 and k=5

```

Example: Sparse U

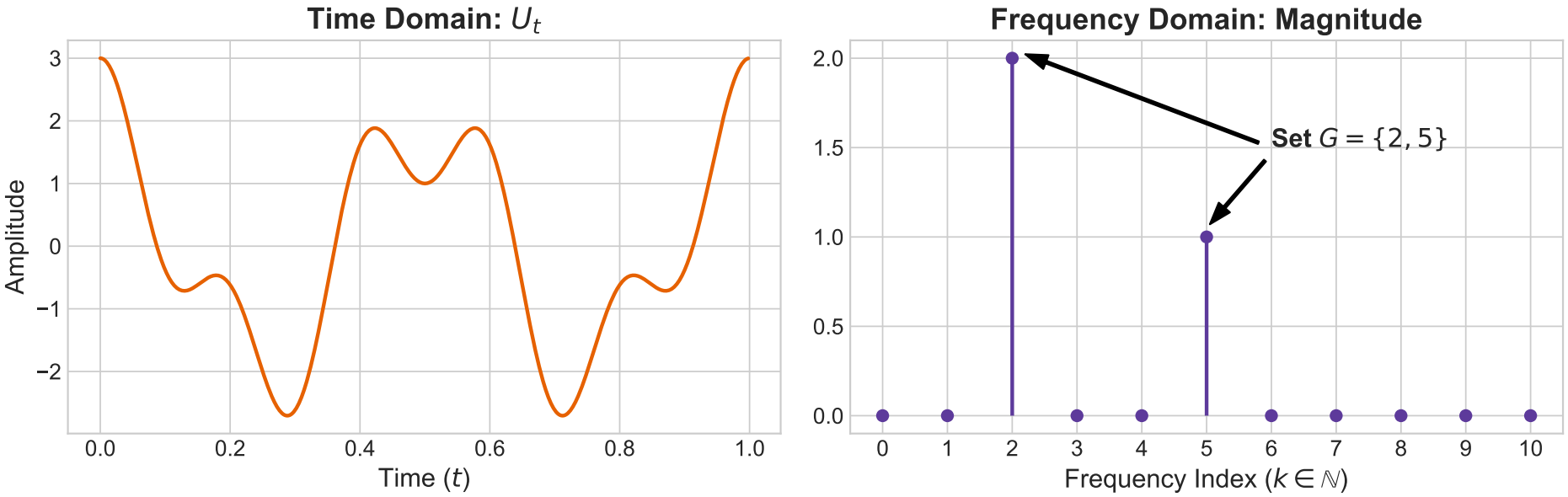

Figure: A sparse confounder U_t in the time domain (left) and frequency domain (right). The non-zero spikes represent the "small" confounded set G.

Frequency Domain Transformation

Transformation $T_k^{\phi,n}$

Define, for all $k \leq n$, the function $T_k^{\phi,n} : \mathcal{S} \to \mathcal{C}$ by:

$$T_k^{\phi,n}(V) := \begin{bmatrix} \frac{1}{n}\sum_{l=1}^{n}(V_{Tl/n})_1\phi_k(Tl/n) \\ \vdots \\ \frac{1}{n}\sum_{l=1}^{n}(V_{Tl/n})_d\phi_k(Tl/n) \end{bmatrix}$$

- $k \leq n$**:** # needed ind. basis functions $\leq$ n (# data points)
- $\sum_t V_t\phi_k(t) \cdot (T/n)$**:** discrete approximation of $\int_0^T V_t\phi_k(t)dt$ with time step $\Delta t = T/n$, $T = 1$
- **DCT:** for the cosine basis, this corresponds to Discrete Cosine Transform

Frequency Domain Transformation - Outlier Problem

The Deconfounded View

Using the linearity of $T_k^{\phi,n}$, it holds for all $k \leq n$:

$$\begin{aligned} T_k^{\phi,n}(Y) &= T_k^{\phi,n}(X)^\top \beta + T_k^{\phi,n}(U) + T_k^{\phi,n}(\eta) \\ &\overset{\text{a.s.}}{=} \begin{cases} T_k^{\phi,n}(X)^\top \beta + T_k^{\phi,n}(U) + T_k^{\phi,n}(\eta) & \text{if } k \in G, \\ T_k^{\phi,n}(X)^\top \beta + T_k^{\phi,n}(\eta) & \text{if } k \notin G. \end{cases} \end{aligned} \tag{2}$$

Remark: The Outlier Problem

If $k \in G$, $T_k^{\phi,n}(U)$ acts as an **adversarial outlier**. If $k \notin G$, the confounder term is zero, leaving an unconfounded linear relationship.

Outlier Problem

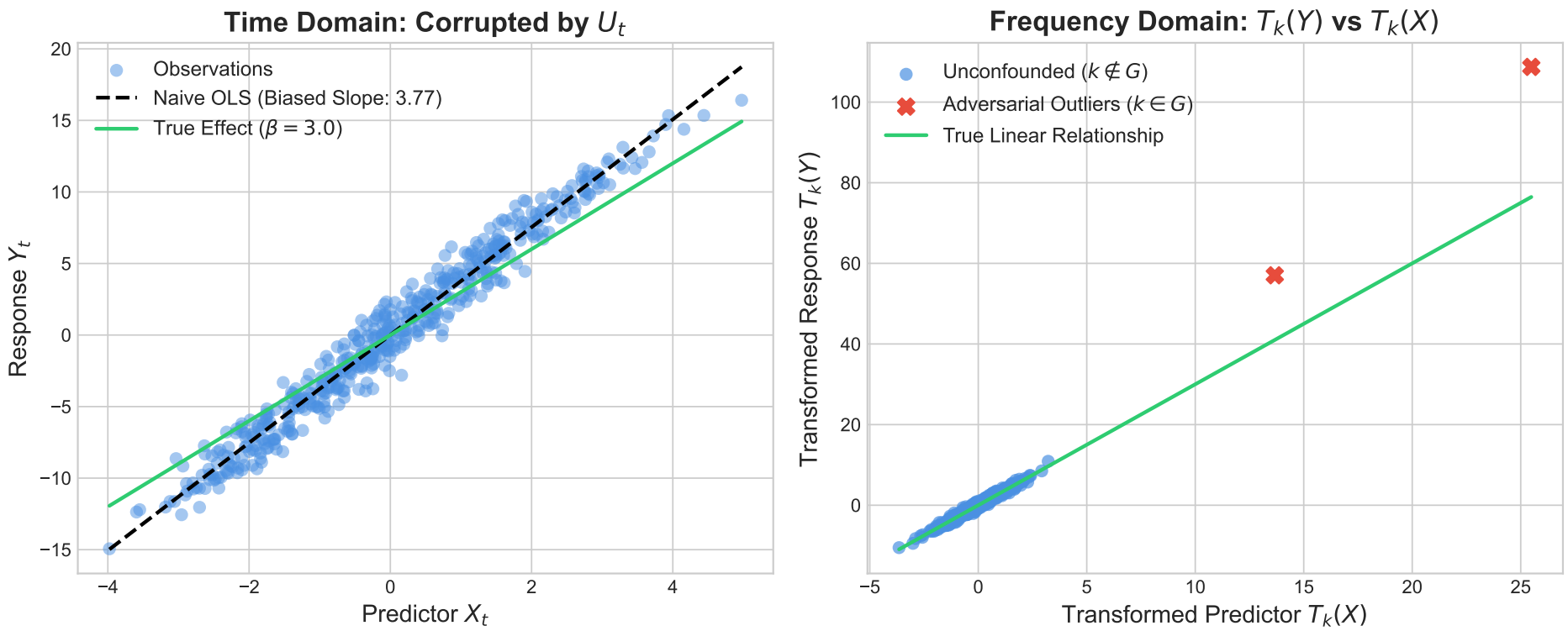

Figure: **Left:** In the time domain, the hidden confounder U_t corrupts X_t and the estimation of β via OLS. **Right:** Applying DCT $T_k^{\phi,n}$ isolates the confounder into the sparse set G. The goal becomes identifying these isolated outliers and estimating the unconfounded relationship β (via Robust Regression).

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The DeCoR Algorithm

Algorithm 1: Deconfounding via Robust Regression

Require: $X^n \in \mathbb{R}^{n \times d}$, $Y^n \in \mathbb{R}^n$, orthonormal basis ϕ, robust linear regression algorithm $\mathcal{A}$

$X^n_\phi \leftarrow (T_1^{\phi,n}(X), \ldots, T_n^{\phi,n}(X))^\top$ ▷ Translate to frequency domain
$Y^n_\phi \leftarrow (T_1^{\phi,n}(Y), \ldots, T_n^{\phi,n}(Y))^\top$
$\hat{\beta}^{\phi,n}_{\text{DeCoR}} \leftarrow \mathcal{A}(X^n_\phi, Y^n_\phi)$ ▷ robust regression
return $\hat{\beta}^{\phi,n}_{\text{DeCoR}}$

Insights:

- **Modular Design:** Complexity shifted to the transformation step ($T^{\phi,n}$).
- **Flexible basis:** Requires only an orthonormal basis ϕ (e.g., Cosine).
- **Plug-and-play solver:** Uses any standard robust regression algorithm $\mathcal{A}$.

The Torrent Algorithm [1]

Algorithm 3: Torrent (Bhatia et al., 2015)

Require $X \in \mathbb{R}^{n \times d}$, $Y \in \mathbb{R}^n$, $a \in \{1, \ldots, n\}$

$S_0 \leftarrow \{1, \ldots, n\}, e \leftarrow Y, \text{err} \leftarrow \infty, t \leftarrow 0$

while $\|e\|_2 < \text{err}$ **do**

  $t \leftarrow t + 1$

  $\text{err} \leftarrow \|e\|_2$

  $\hat{\beta}^t_{\text{Tor}} \leftarrow \hat{\beta}^{S_{t-1}}_{\text{OLS}}(X, Y)$ ▷ Fit OLS on subset S_{t-1} from prev. iteration

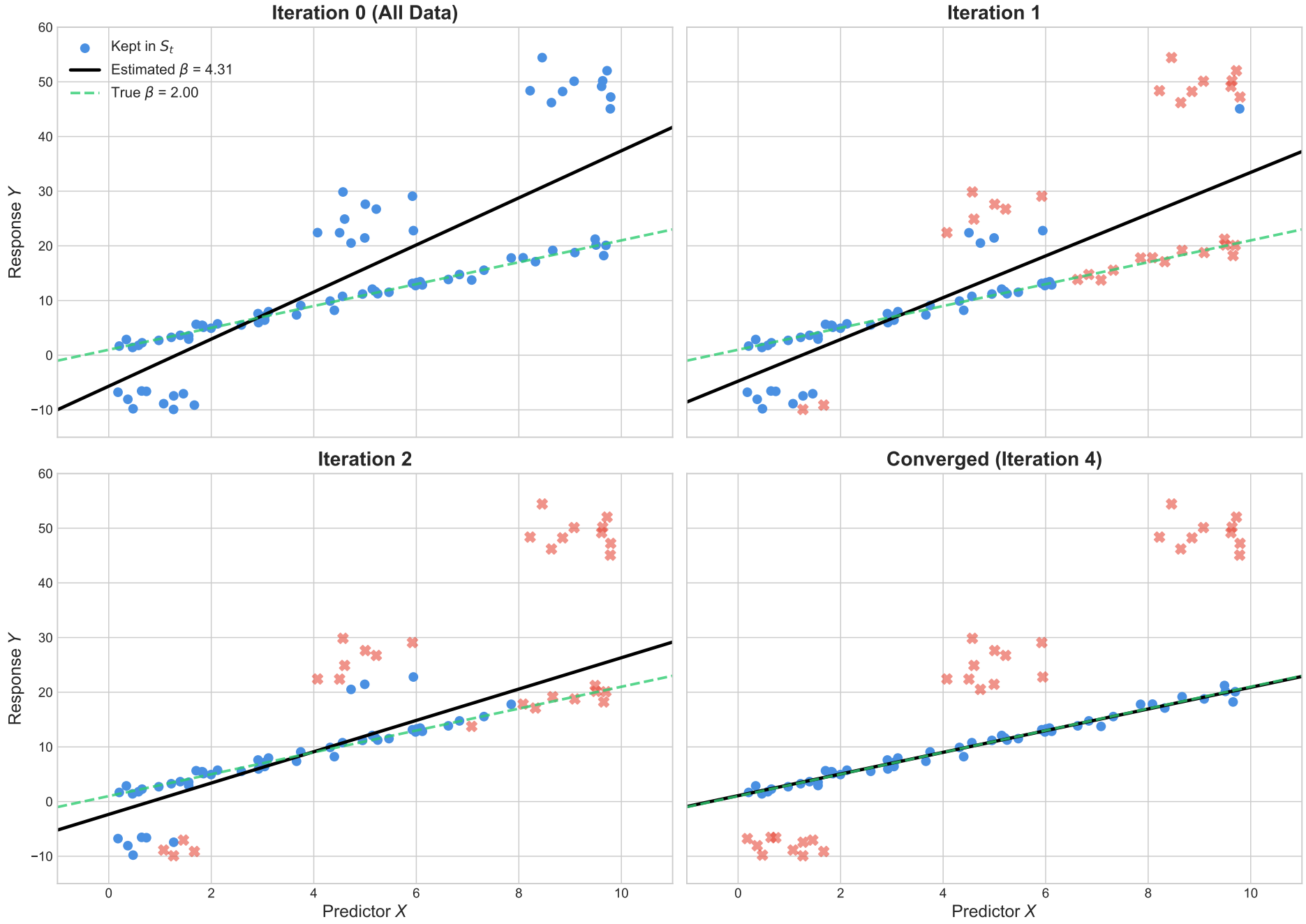
  $v \leftarrow |Y - X\hat{\beta}^t_{\text{Tor}}|$ ▷ Calc residuals for ALL data points

  $S_t \leftarrow \text{HT}(v, a)$ ▷ Keep a many points with smallest residuals

  $e \leftarrow |Y_{S_t} - X_{S_t}\hat{\beta}^t_{\text{Tor}}|$ ▷ Measure error on the new clean subset S_t

end while

return $\hat{\beta}^{n,a}_{\text{Tor}} := \hat{\beta}^t_{\text{Tor}}$


Iteration 0 (All Data)
Iteration 1
Iteration 2
Converged (Iteration 4)
Kept in S_t
Estimated β = 4.31
True β = 2.00
Response Y
Predictor X
60
50
40
30
20
10
0
−10
0
2
4
6
8
10

Empirical Convergence of Torrent

n	**mean**	**min**	**max**
10	2.42	2	3
100	5.14	3	8
1000	8.26	4	13

Convergence Speed

Even for a large sample size of $n = 1000$, Torrent converges in < 15 **iterations**.

Experimental setting: X and U are band-limited processes, theshold param. $a = 0.7$, elements of G_n chosen uniformly at random with probability 0.25 [2].

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Setting the Stage for Proofs for Torrent

Key Assumptions

1. **Setting 2:** Let $n \geq d$. We observe $X \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^n$ defined by:

$$Y := X\beta + \epsilon + o$$

   where $\epsilon \in \mathbb{R}^n$ is noise and $o \in \mathbb{R}^n$ are sparse outliers supported on G_n (i.e., $o_i = 0$ for $i \notin G_n$). We call G_n^c the **inliers**.
2. **Bounded Outliers G_n:** # confounded frequencies $|G_n| \leq c_n$.
3. **Mistakes-Stability-Ratio:** As $n \to \infty$, it holds with high probability that:

$$\eta := \max_{S \text{ s.t. } |S| = n - c_n} \frac{||X_{V(S)}||_2}{\sqrt{\lambda_{\min}(X_S^\top X_S)}} < \frac{1}{\sqrt{2}}$$

   where $S \subseteq \{1, ..., n\}$ and $V(S) = (S \cup G_n^c) \setminus (G_n^c \cap S)$ is the symmetric difference between S and G_n^c.

Intuitions:

- Should η better be nearly 0 or close to our upper bound?
- What happens if X_S is singular for some S?

Proofs - Torrent (deterministic)

Guarantee of Torrent (Theorem 3)

Let S_t be the final estimated subset of Torrent on data X, Y using threshold parameter $a_n := n - c_n$. Then

$$\|\hat{\beta}^{n,a_n}_{\text{Tor}} - \beta\|_2 \leq \text{Err}_{\text{OLS}} + \frac{\sqrt{2}\|X_{V(S_t)}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon_{V(S_t)}\|_2}{\sqrt{\lambda_{\min}(X_{S_t}^\top X_{S_t})}(1 - \sqrt{2}\eta)}.$$

with $\text{Err}_{\text{OLS}} = \|(X_{S_t}^\top X_{S_t})^{-1} X_{S_t}^\top \epsilon_{S_t}\|_2$ *being the baseline error of the clean subset.*

Intuitions:

- What happens in case of no outliers at all?
- What happens if $\eta \approx \frac{1}{\sqrt{2}}$ or X_{S_t} is singular?
- Note the multiplicative/additive effects of outliers in X and Y

Detailed Proof - Theorem 3

Assume $X_{S_{t-1}}^\top X_{S_{t-1}}$ is invertible.

1. Optimality Bound (Eq. 27)

We use the optimality of the model $\hat{\beta}_{\text{Tor}}^t$ on the active set S_{t-1}:

$$\begin{aligned}
\|\hat{\beta}_{\text{Tor}}^t - \beta\|_2 &= \left\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1}(X_{S_{t-1}}^\top \epsilon_{S_{t-1}} + X_{S_{t-1}}^\top o_{S_{t-1}})\right\|_2 \\
&\leq \left\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1} X_{S_{t-1}}^\top \epsilon_{S_{t-1}}\right\|_2 \\
&\quad + \left\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1} X_{S_{t-1}}^\top\right\|_2 \|o_{S_{t-1}}\|_2
\end{aligned}$$

2. Hard Thresholding Guarantee (Eq. 28)

Following the HT-step and using relation $|G_n| \leq c_n \implies |G_n^c| \geq n - c_n$:

$$\begin{aligned}
\|X_{S_t}(\beta - \hat{\beta}_{\text{Tor}}^t) + \epsilon_{S_t} + o_{S_t}\|_2^2 &= \|Y_{S_t} - X_{S_t}\hat{\beta}_{\text{Tor}}^t\|_2^2 \\
&\leq \|Y_{G_n^c} - X_{G_n^c}\hat{\beta}_{\text{Tor}}^t\|_2^2 \\
&= \|X_{G_n^c}(\beta - \hat{\beta}_{\text{Tor}}^t) + \epsilon_{G_n^c}\|_2^2
\end{aligned}$$

Bounding the Outliers $\|o_{S_t}\|_2$:
Let $H_t := S_t \setminus G_n^c$ and $M_t := G_n^c \setminus S_t$. We know from above:

$$\|X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{H_t} + o_{H_t}\|_2 \leq \|X_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{M_t}\|_2$$

Let $H_t \cup M_t = V(S_t) =: V_t$. We can then expand by triangle inequality:

3. Outlier Expansion (Eq. 29)

$$\begin{aligned}
\|o_{S_t}\|_2 &\leq \|X_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon_{H_t}\|_2 + \|\epsilon_{M_t}\|_2 \\
&\leq \sqrt{2}\,\|X_{V_t}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon_{V_t}\|_2 \\
&\leq \sqrt{2}\,\|X_{V_t}\|_2 \Big(\Big\|(X^\top_{S_{t-1}} X_{S_{t-1}})^{-1} X^\top_{S_{t-1}} \epsilon_{S_{t-1}}\Big\|_2 \\
&\qquad + \Big\|(X^\top_{S_{t-1}} X_{S_{t-1}})^{-1} X^\top_{S_{t-1}}\Big\|_{2\to 2} \|o_{S_{t-1}}\|_2 \Big) + \sqrt{2}\,\|\epsilon_{V_t}\|_2
\end{aligned}$$

- Add and subtract: $\|o_{S_t}\|_2 = \big\|(X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{H_t} + o_{H_t}) - (X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{H_t})\big\|_2$
- Apply $(|a| + |b|)^2 \leq 2(a^2 + b^2)$ twice to merge H_t and M_t into V_t:
  - For the noise: $(\|\epsilon_{H_t}\|_2, \|\epsilon_{M_t}\|_2) \to \sqrt{2}\|\epsilon_{V_t}\|_2$
  - For the signal: $(\|X_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2, \|X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2) \to \sqrt{2}\|X_{V_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2$
- Observe $\|o_{H_t}\|_2 = \|o_{S_t}\|_2$, recalling $o_i = 0$ for $i \notin G_n$

Let t be the final iteration of Torrent, meaning $S_t = S_{t-1}$.
By assumption:

$$\eta = \max_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S|=a_n} \frac{\|X_{V(S)}\|_2}{\sqrt{\lambda_{\min}(X_S^\top X_S)}} = \max_S \|X_{V(S)}\|_2 \|(X_S^\top X_S)^{-1} X_S^\top\|_2 < \frac{1}{\sqrt{2}}$$

- Recall the matrix norm property: $\|A\|_2 = \sqrt{\lambda_{\max}(A^\top A)} = \sqrt{\lambda_{\max}(AA^\top)}$.
- Set $A := (X_S^\top X_S)^{-1} X_S^\top$ and observe $AA^\top = (X_S^\top X_S)^{-1}$
- $\|(X_S^\top X_S)^{-1} X_S^\top\|_2 = \|A\|_2 = \sqrt{\lambda_{\max}(AA^\top)} = \sqrt{\lambda_{\max}((X_s^\top X_s)^{-1})} = 1/\sqrt{\lambda_{\min}((X_s^\top X_s))}$

4. Applying η to isolate o_{S_t} (Eq. 30)

Distributing terms in Eq. 29 and substituting the η bound:

$$\|o_{S_t}\|_2 \leq \underbrace{\sqrt{2}\|X_{V_t}\|_2 \left\| \left(X_{S_t}^\top X_{S_t}\right)^{-1} X_{S_t}^\top \right\|_2}_{\leq \sqrt{2}\eta} \|o_{S_t}\|_2$$

$$+ \sqrt{2}\|X_{V_t}\|_2 \left\| \left(X_{S_t}^\top X_{S_t}\right)^{-1} X_{S_t}^\top \epsilon_{S_t} \right\|_2 + \sqrt{2}\|\epsilon_{V_t}\|_2$$

$$\implies \|o_{S_t}\|_2 \leq \frac{\sqrt{2}\|X_{V_t}\|_2 \left\|(X_{S_t}^\top X_{S_t})^{-1} X_{S_t}^\top \epsilon_{S_t}\right\|_2 + \sqrt{2}\|\epsilon_{V_t}\|_2}{1 - \sqrt{2}\eta}.$$

Proofs - Torrent (probabilistic)

Define the square root of minimum eigenvalue of all possible clean subsets as

$$\lambda_{a_n}(X) := \min_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S| = a_n} \sqrt{\lambda_{\min}(X_S^\top X_S)}$$

Guarantee of Torrent with sub-Gaussian noise (Corollary 4)

Let ϵ be i.i.d. zero-mean sub-Gaussian noise with variance proxy σ^2 and assume $c_n < n/2$. Then, there exists a constant $K > 0$ s.t. $\forall\ \delta > 0$ with probability $\geq 1 - \delta$:

$$\|\hat{\beta}_{\text{Tor}}^{n,a_n} - \beta\|_2 \leq \frac{\sigma}{\lambda_{a_n}(X)} \left(1 + \frac{1}{1 - \sqrt{2}\eta}\right) \left(\sqrt{d} + \sqrt{2c_n} L_{n,\delta}\right) + \frac{2\sigma\sqrt{c_n}\,(1 + L_{n,\delta})}{\lambda_{a_n}(X)(1 - \sqrt{2}\eta)}.$$

with $L_{n,\delta} := \sqrt{K \log(2en/c_n\delta)}$ representing the logarithmic search penalty.

Insights:

- What happens if $c_n = 0$? Compare it to standard OLS.
- If the rows of X are i.i.d. standard Gaussian random vectors (i.e. $\lambda_{a_n}(X) \in \Omega_{\mathbb{P}}(\sqrt{n})$ [1]) and $c_n \in o(n/\log(n))$, then $\hat{\beta}_{\text{Tor}}^{n,a_n}$ is consistent.

Convergence properties of DecoR-Tor (Theorem 6)

Let c_n be a known sequence of non-negative natural numbers such that $|G_n| \leq c_n$ and let Assumption 2 in [1] be satisfied. Assume that for all $\delta > 0$ there exists $\bar{n} \in \mathbb{N}$ such that for all $n \geq \bar{n}$ it holds that

$$\mathbb{P}\left[\max_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S| = n - c_n} \frac{\|(X_\phi^n)_{V(S)}\|_2}{\sqrt{\lambda_{\min}\left((X_\phi^n)_S^\top (X_\phi^n)_S\right)}} < \frac{1}{\sqrt{2}}\right] \geq 1 - \delta.$$

If DecoR is executed with Torrent and the sequence of threshold parameters $a_n = n - c_n$, then

$$\|\hat{\beta}_{\text{DecoR}}^{\phi,n} - \beta\|_2 \in \mathcal{O}_{\mathbb{P}}\left(\sigma\sqrt{\frac{c_n \log(n/c_n)}{n}}\right).$$

Insights:

- $\hat{\beta}_{\text{DecoR}}^{\phi,n}$ is a consistent estimator of β, if $\sigma = 0$ or $c_n \in o(n)$
- if noise is non-zero and c_n is asymptotically proportional to n, then the estimation error is bounded by σ, up to constant factors.

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Synthetic Data - Data Generation

Recall:

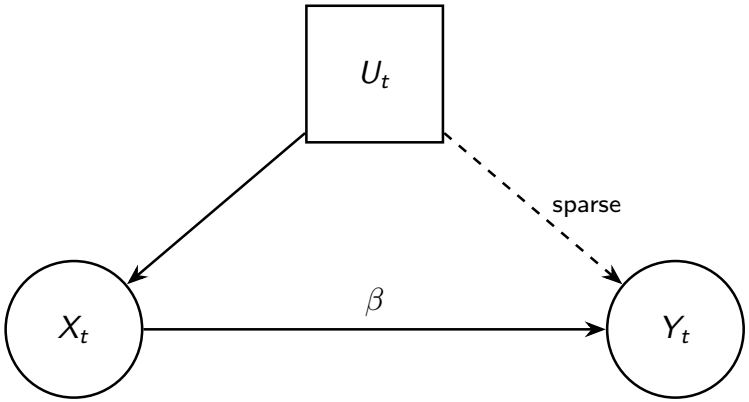


Figure: Causal Graph of X and Y with hidden confounder U [3]

We set in our one-dimensional scenario:

- $U := \text{AR}(1) + \epsilon_u$ with AR(1) := Ornstein-Uhlenbeck process and noise $\epsilon_u \sim N(0, 1)$
- $U_{sparse} = \text{sparsify}(U, |G_n| = 0.25 \cdot n, \text{cosine-basis})$
- $X := \text{AR}(2) + U_{sparse} + \epsilon_x$ with $\epsilon_x \sim N(0, 1)$
- $Y := 3X + 10 \cdot U_{spasre} + \epsilon_y$ with $\epsilon_y \sim N(0, 1)$

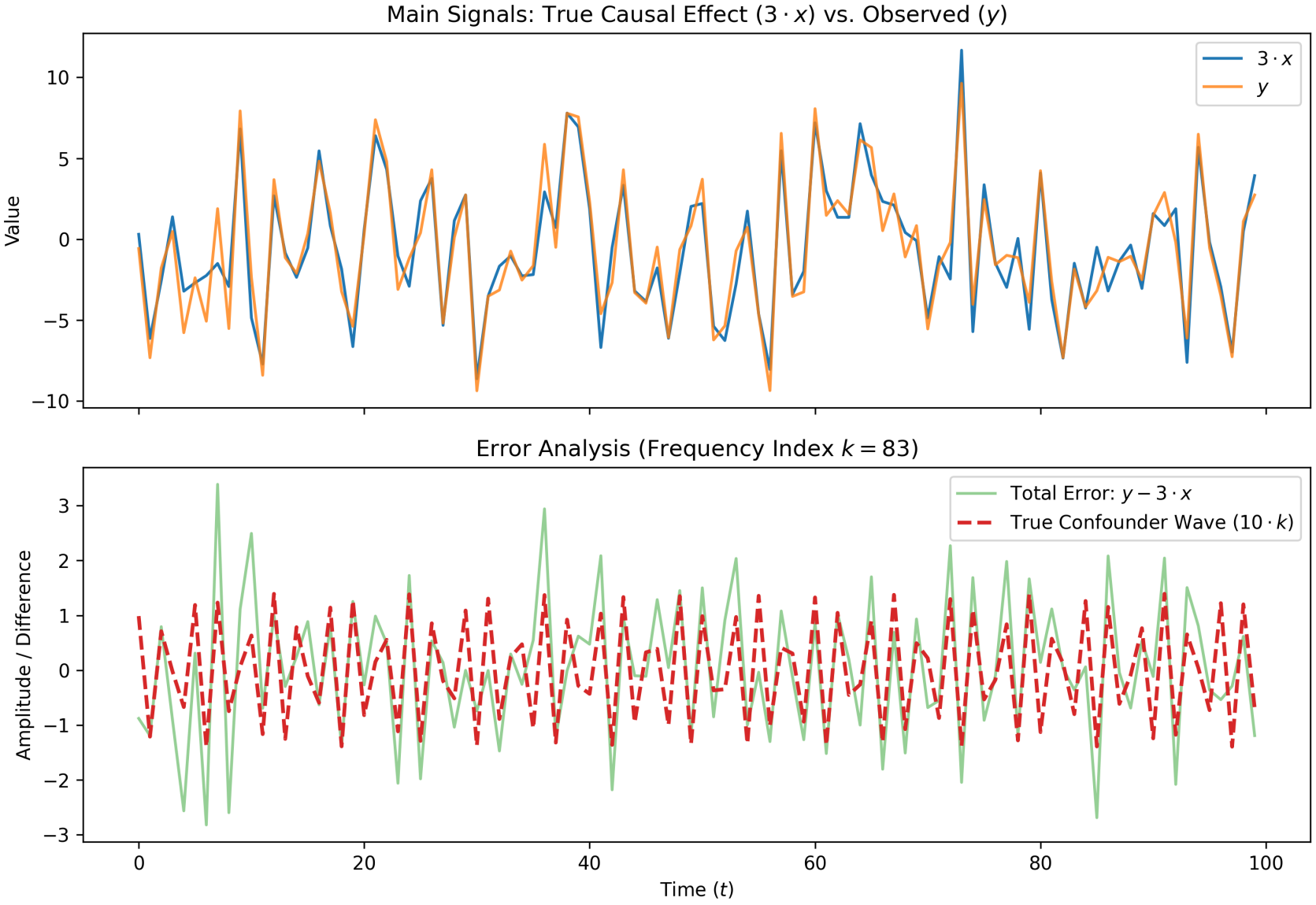


Figure: Visualization of the true causal signal $(3 \cdot X)$, observed data (Y), and the isolated sparse confounder (U_{sparse}) with $n = 100$, $G_n = \{k\}$

Synthetic Data - Experimental Results

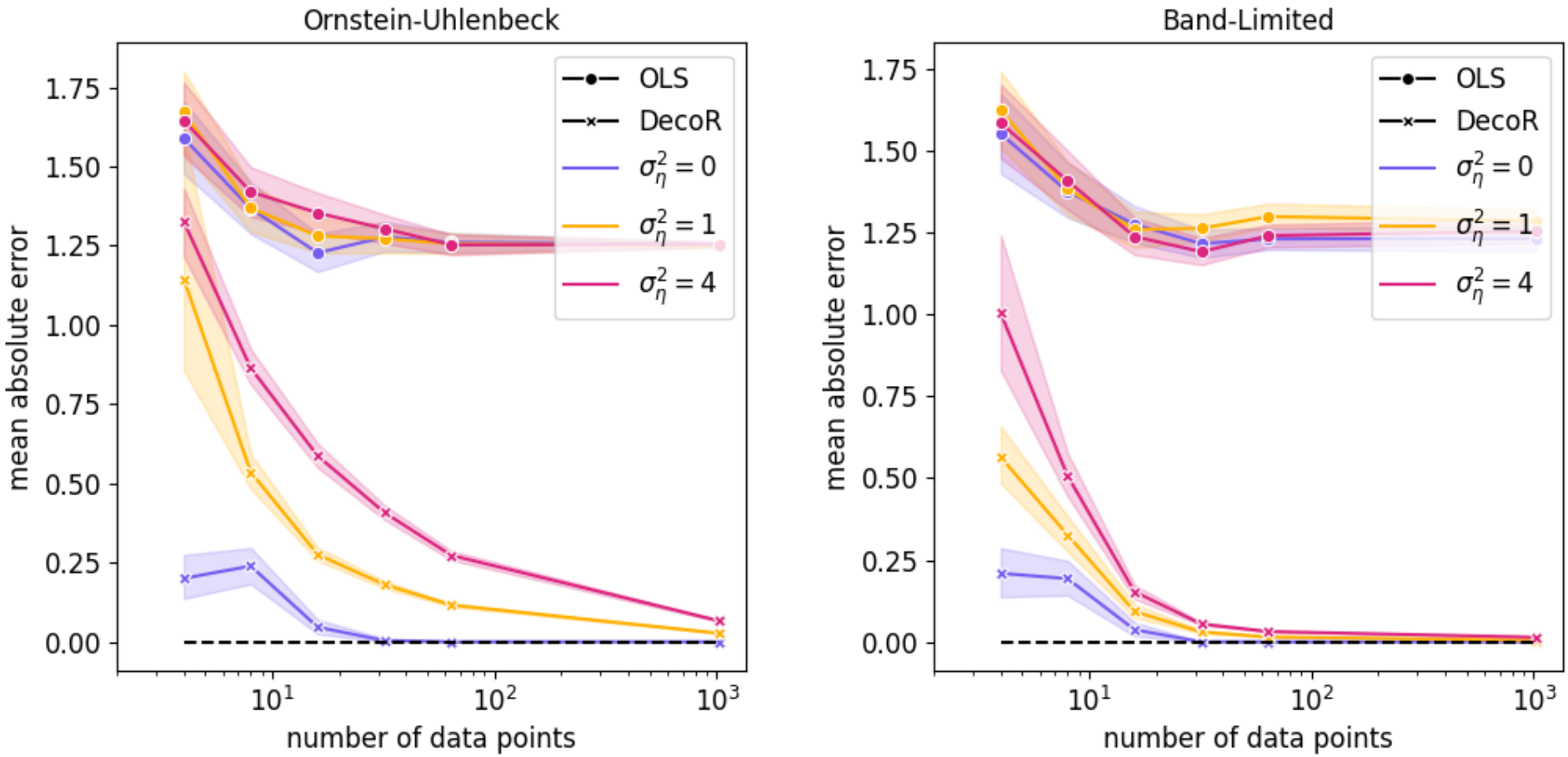
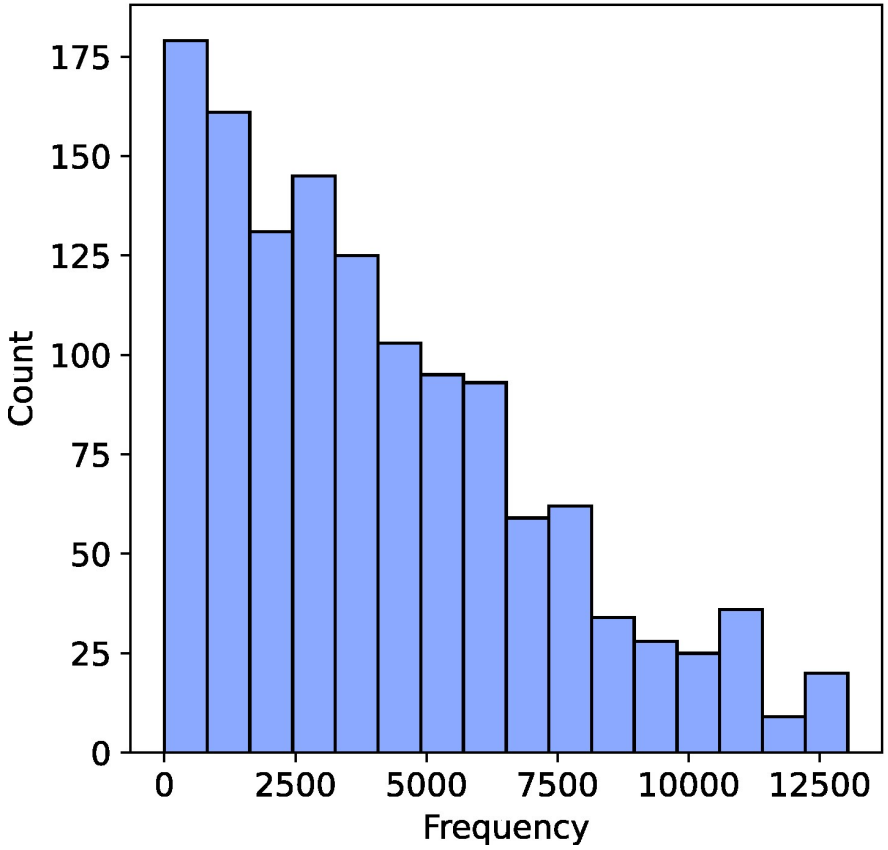
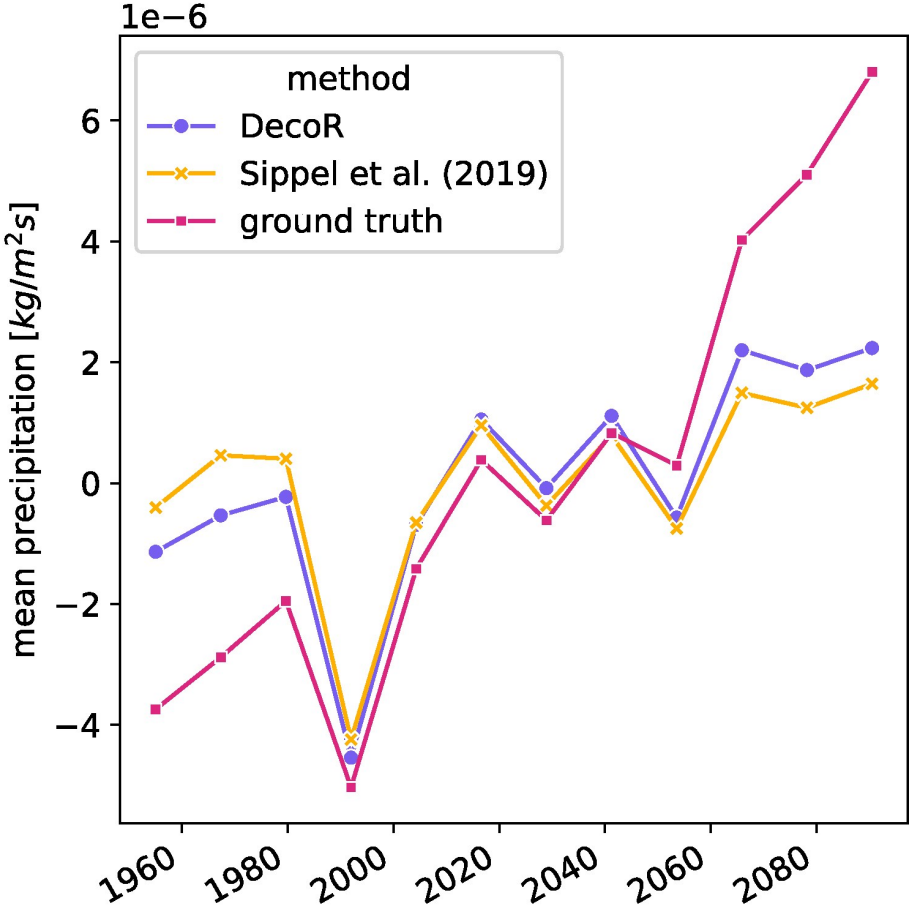


Figure: Comparison of DecoR performance against OLS with $|G_n| = 0.25$ and cosine-basis. In the right figure, band B of size 50 is used.

Application: Earth System Science





Findings: DeCoR automatically identifies and filters out the slow, low-frequency components associated with climate change, accurately isolating the short-term "weather" signal β.

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Conclusion

- **The Problem:** Unobserved confounders severely bias standard regression in time series.
- **The DeCoR Insight:** By moving to the frequency domain, we exploit the sparsity of confounders.
- **The Method:** Treating confounded frequencies as adversarial outliers allows robust regression (like Torrent) to recover the true causal effect.

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Uncovering the forced climate response from a single ensemble member using statistical learning.
Journal of Climate, 32(17):5677 – 5699, 2019.

Backup

Convergence properties of DecoR-Tor (Theorem 6) - shortened

$$\|\hat{\beta}^{\phi,n}_{\text{DecoR}} - \beta\|_2 \in \mathcal{O}_{\mathbb{P}}\left(\sigma\sqrt{\frac{c_n \log(n/c_n)}{n}}\right).$$

Proof ideas:

- By some assumptions, note that $\sqrt{\lambda_{\min}\left((X^n_\phi)^\top_S (X^n_\phi)_S\right)} \geq c'$ with high probability.
- each component of $\eta^n_\phi = \{T^{\phi,n}_k(\eta)\}_{k\leq n} = \frac{1}{n}\sum_{l=1}^n (\eta_{Tl/n})_k \cdot \phi_k(Tl/n)$ has (by factor n) reduced variance $\bar{\sigma}^2 = \sigma^2/n$
- Use **Corollary 4** and derive Big-$\mathcal{O}$ result:

$$\|\hat{\beta}^{n,a_n}_{\text{Tor}} - \beta\|_2 \leq \frac{\sigma}{\lambda_{a_n}(X)}\left(1 + \frac{1}{1-\sqrt{2}\eta}\right)\left(\sqrt{d} + \sqrt{2c_n K \log\left(2en/c_n\delta\right)}\right) + \frac{2\sigma\sqrt{c_n}\left(1 + \sqrt{K\log\left(2en/c_n\delta\right)}\right)}{\lambda_{a_n}(X)(1-\sqrt{2}\eta)}.$$